The web we use today was not inevitable, it was shaped by what its builders believed, the technology they made, and the communities that set the rules.
We're now living through a similar transformational period like the one of the beginning of the web in which we can have a saying about how AI will shape our lives.
In this talk I motivate the path I believe is the most effective towards building "A beautiful future for the Web commons".

A beautiful future For the web Commons

Authors

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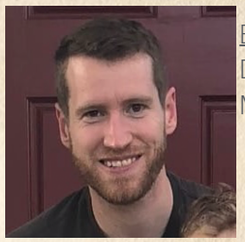
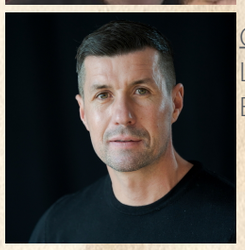


Brian Grinstead
Distinguished engineer at Mozilla



Christoph Kerschbaumer
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This work is a collaboration with the center for human compatible AI at Berkeley and Mozilla and my participation is funded by the Foresight Institute, so thank you for that!

The hierarchy beneath the norms, the rules and the laws

- Philosophy
  - The underlying commitments
- Technology
  - The layer that brings ideas into existence
- Communities
  - The adaption of philosophy and technology in society

We believe that the Web today, at both the consumer and the regulatory level, was the result of a chain of complex events that couldn't really be predicted but was certainly influenced, and even bounded, by the underlying its building blocks. We argue that philosophy, technology, and our communities are, in this order, responsible for the ways the social and political layer of the Web is formed.

The hierarchy beneath the norms, the rules and the laws

- Philosophy
- Technology
- Communities

Our claim is that if each layer is built on the right values…

The hierarchy beneath the norms, the rules and the laws

- Philosophy
- Technology
- Communities
- The norms, rules and laws that ensure an enjoyable commons

…then a Web commons with enjoyable norms, fair rules, and privacy-assuring laws will be the beautiful web we want to be in.

The hierarchy beneath the norms, the rules and the laws

- Philosophy
- Technology
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- The norms, rules and laws that ensure an enjoyable commons

However of course if these are built in shaky grounds then the commons might fall prey to…

The hierarchy beneath the norms, the rules and the laws

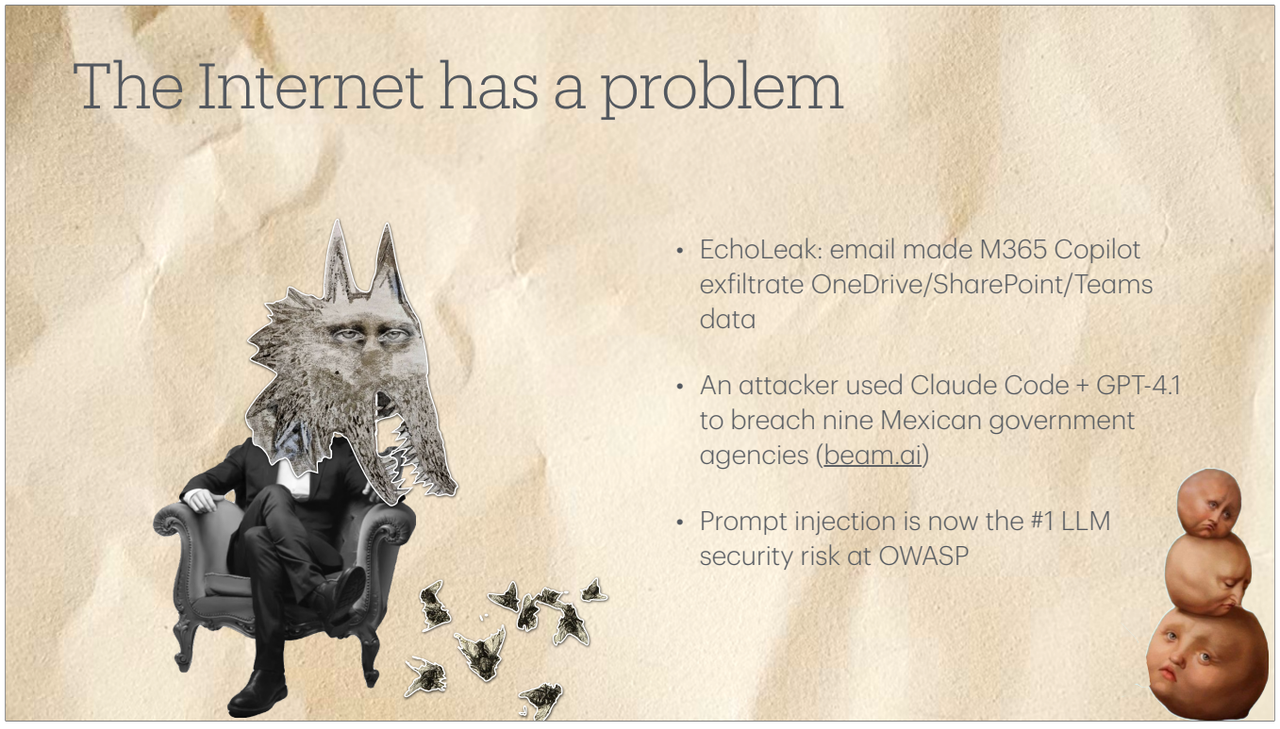
- Philosophy
- Technology
- Communities
- The norms, rules and laws that ensure an enjoyable commons

…exploitation by those who have divergent or actively harm incentives to the population using the the Web Commons. In this sense, the Web Commons is much like any other commons we usually take part in.

The Internet has a problem

- **Automated requests are 57.5%** of HTML web traffic, versus **42.5% from humans** (Forbes 2026)
- **Agentic AI traffic grew 7,851%** year over year
  OpenAI alone generates ~69% of all AI bot traffic; Meta ~16%, Anthropic ~11%, everyone else under 5%
- **Attackers hijacking** accounts after login **quadrupled** (402,000 attempts per organization*)

Agentic AI traffic in the web surpassed the traffic generated by humans this year. The large majority of this traffic is not generated by users but by big corporations like OpenAI, Meta and Anthropic. Attacks that can exploit agent's vulnerabilities like prompt injection have quadrupled in the last year.



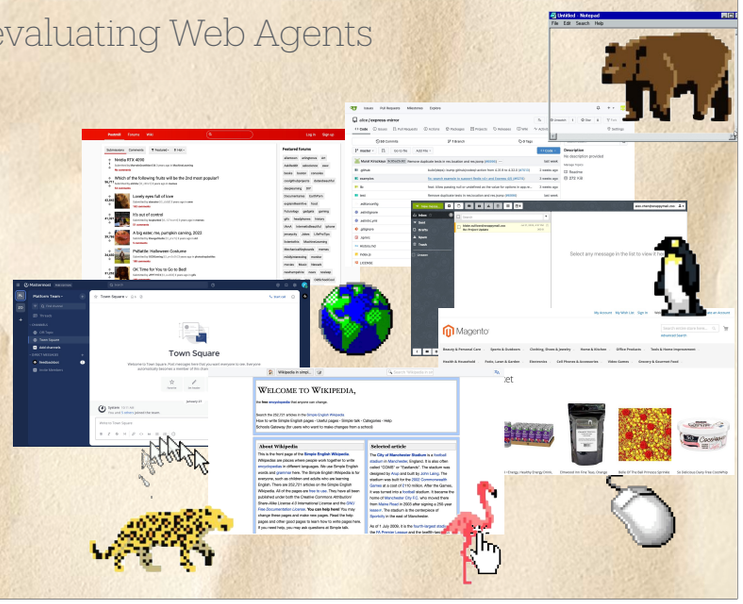
Agentic AI has been used to carry on through a number of attacks both in real and sandboxed environments, and we expect these numbers to increase. The core reason why these systems aren’t being widely used today is because prompt injection remains an unsolved problem.

I’d like to present the parts of the solution we’re working on in order to improve the state of the agentic web and I’d also like to motivate my workshop proposal which is to draft a manifesto for the Web Commons.

From the Wild Web to the Zoo

A realistic environment for evaluating Web Agents

- Real web protocols
- Interconnected apps
- Fully deterministic
- Open source



One of the main challenges of developing web agentic technology has been the lack of a deterministic sandbox environment that captures the messiness of the web. So, before solving any other issues we've built Zoo.

<click> Zoo is a deterministic web that lives on your own machine. It's a simulated internet of real, interconnected applications like email, e-commerce, git, chat, reddit and others. Zoo runs behind real web protocols, real DNS, and real TLS, so an agent navigates it exactly as it would the open web, sending emails to one another, interfering on each other's workflow and so on.
Unlike the open web, Zoo gives you two extra things: full visibility into the backend state of every service, and deterministic resets, so every experiment can start from identical conditions.

We then built an evaluation layer on top of Zoo.
While Zoo provides the environment, the evaluation layer provides the tools to construct realistic scenarios inside it and measure how agents perform.
It's organized around three ideas: universes, tasks and scenes.
<click>
Universes, define the world an agent acts, it contains personas, other agents and websites seeded with data.
<click>
Tasks define what the agent must accomplish and how success is measured. And because Zoo exposes full backend state, scoring an agent isn't limited to reading the its output, a verifier can check ground truth directly, by querying a database or running custom logic.
<click>
And **scenes**, which bring the environment to life. It allows users to create deterministic events that fire mid-task allowing you to stage reproducible dynamic attacks like a prompt injection that is triggered by a JavaScript action for example.
Zoo is harness and model agnostic which means you can use any kind of scaffolding system and model to have your agent interacting with the web browser.

The second part of this project which is still under development is a software that allows agent to navigate the web while being verifiably safe against prompt injections.
To do that, we're building a system that couples the language model with non-ML components and native web-standard signals, so the guarantees don't depend on the model behaving.
<click>
In our system, the agent never has access to sensitive like your address, or credit card, it requests an encrypted deterministic layer to take over and deal with your sensitive data.
<click>
This layer is represented by a finite-state machine and it has to be implemented by a trusted source with a programming language. It is responsible for performing a series of checks that guarantees that the current state matches user’s requirements. We do that by checking variables like website’s domain, whether the connection is TLS-secured, device & browser fingerprint, and other network intelligence measures. We also confirm whether the environment checks out with custom user set policies. The result of is an agent that's useful and verifiably safe.

<click>
We’re also interested in what a soft layer of guarantees could offer. This approach can offer more flexibility than a deterministic layer. For example, we can steer models against prompt injection patterns or gate them against providing certain information given different levels of privacy.

There are, of course, limits for these two kinds of approach approach, because the Web is messy, or a “Zoo” , the first approach might hinder the ability of agents to be fully agentic and the second might just not be sufficiently safe.

<click>
This tradeoff could be significantly minimized if we establish universal protocols for agent behavior on the web. For example, we could have a safe API similar to the one used by bank systems in which the finite-state machine system I just described interacts directly with the entity you’re trying to establish a transaction, diminishing the variability and the input of agents in such interactions.

We need to have organizations like W3C (that is responsible for setting standards for the web) to develop protocols that guarantee user privacy and safety.

These protocols are lobbied by people’s interests, though often these people aren’t aligned with our interests.

<Possible end>

That’s why I believe community is so important. Philosophy and technology can only build the foundation layers but people will establish the norms and laws that rule the web.

The early web has shown us how communities can be pivotal. Many of the Web’s positive characteristics exist because communities fought for them. Some examples of this are free, universal encryption by Let’s Encrypt and net neutrality that establishes that Web traffic can’t be prioritized. Others wins were championed by heroes like Aaron Swartz and Elbakyan, that enabled free knowledge sharing through platforms like SciHub.

Many of the webs faults happened because communities lost their fight or didn’t give them enough thought. Some examples here are the walled gardens created by big tech, government surveillance and protocols owned by the few.
To name one example, in the early 2010s you could activate something called DNT (Do not track) and send this information so websites wouldn’t track you across websites. This was by no means enforced and it wasn’t adopted because it went directly against the incentives of the ad companies. It’s not a simple answer because the landscape of stakeholders in the web is extremely diverse.

We as researches, technologists and citizens of the Web, have the power to shape many of the layers that the agentic AI will stand on.

This is what I had to say and I’d be happy to hear from you what does it take to co-create norms and protocols that not only will be adopted by the community but cherished as well.

Thank you

That’s why I believe community is so important. Philosophy and technology can only build the foundation layers but people will establish the norms and laws that rule the web.

To begin this conversation here I’d like to propose we build a manifesto for user first agents.
The underlying assumptions of this manifesto is that **AI can be a transformative force for human development** and we carry on our work with the Good Faith that we can build a future in which that is true.
First, is it possible to build realistically useful systems with agents having only c**ompartmentalized access to the data**, or even better, no access at all? How should we design the protocols that control privacy settings on the web to users?
Second, **Agent interoperability:** I believe system interoperability, that is, open protocols any provider can plug into instead of proprietary silos is good, but how can we make sure that these protocols are going to be adopted by stakeholders?
Third, **Agent allegiance:** Should agents respond to user’s interests primarily? Maybe here too we’d want to have guarantees that the models are well aligned and they won’t for example, engage in cyber attacks.
Fourth, should we have LLM systems and scaffolding systems to be completely open to user choice? Or should we gate models to attend a minimal measure of safety and security in order to allow them to be used in the Web?
And finally, who should write the protocols that govern how agents act on our behalf? Should it be a fully democratic process or do we want specialists taking over?

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